

# Abanoub AbdelMessieh

Computer Engineer | Firmware Developer | RTOS, VoIP, CMSIS, CRA, IoT

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## PROFESSIONAL SUMMARY

Embedded Firmware & Computer Engineer with 3+ years of professional R&D experience developing real-time firmware for constrained ARM Cortex-M systems. Strong focus on FreeRTOS, bare-metal drivers, VoIP audio pipelines, secure bootloaders, and low-level Ethernet systems. Experienced in translating Linux/POSIX networking behavior into deterministic MCU firmware under tight memory, timing, and flash constraints.

## CORE SKILLS

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|----------------------------------|-----------------------------------------------------------------------------------------|
| <b>Programming</b>               | C, ARMv7, RISC-V, Python, C++                                                           |
| <b>RTOS &amp; Firmware</b>       | FreeRTOS, bare-metal development, Bootloaders, FOTA, Zephyr                             |
| <b>Networking &amp; I/O</b>      | OPEN Alliance 10BASE-T1S, raw Ethernet, LwIP, MQTT, SIP/SDP/RTP, SPI/I2S DMA, UART, I2C |
| <b>Security &amp; Compliance</b> | Cyber Resilience Act (CRA), EC-PKI, TLS, Secure Boot                                    |
| <b>Tools &amp; DevOps</b>        | ARM CMSIS Toolbox, STM32Cube, Keil, GCC/GDB, QEMU, J-Link/Ozone, Docker, Nix            |

## PROFESSIONAL EXPERIENCE

### Urmet S.p.A. - Turin, TO, Italy

Nov 2022 – Present

Embedded Software Designer, Technology Innovation Division – Turin, Italy

- Researched and evaluated various embedded technologies to drive modern R&D practices and break legacy development habits; architected and delivered reusable, platform-agnostic firmware components across group subsidiaries.
- Conducted product security risk assessments under the EU Cyber Resilience Act (CRA) for Class I smart home devices with integrated security functions.
- Engineered a bare-metal Baresip VoIP client (SIP/SDP) on STM32/N32 Cortex-M4, optimizing real-time G.722 DSP assembly routines (reducing loop execution from 4 ms to 3 ms) to stabilize Acoustic Echo Cancellation blueprints.
- Architected a 32 KB secure Layer-2 FOTA bootloader sandbox utilizing raw MAC frames and EC-PKI verification, delivered to the product team for production integration.
- Refactored Microchip 10BASE-T1S TC6 MACPHY drivers to run asynchronously via SPI DMA interrupts, eliminating polling overhead.

### Adgenera - Turin, TO, Italy

Sep 2022 – Oct 2022

Industrial Automation Intern – Curricular Internship

- Contributed to the Zurich Airport baggage handling system renewal project, configuring high-reliability Siemens SIMATIC S7-1500 PLC control nodes.
- Implemented and tested automation logic in Structured Control Language (SCL) using TIA Portal v17, including simulation checks for edge-case sorting behavior.

## SELECTED PROJECTS

For detailed system architectures, interactive sequence diagrams, and signal traces, please visit my portfolio website: [abanoubn.github.io](https://abanoubn.github.io)

### Heterogeneous Secure FOTA & MQTT Sandbox

R&D simulation environment validating secure firmware updates (FOTA) on a hybrid fleet of physical (STM32 Nucleo) and emulated (QEMU Cortex-M4) IoT nodes. Implemented TLS 1.3 mTLS over Mosquitto using coreMQTT/MbedTLS, and built an asyncio-based AWS IoT Jobs emulator to stream CBOR-encoded firmware packages.

### High-Performance Deterministic IP Audio Engine

Real-time VoIP/audio firmware on Cortex-M4 using FreeRTOS, LwIP, SIP/RTP, G.722, I2S DMA, and SPI DMA Ethernet. Focused on deterministic packet turnaround, low-jitter audio streams, and constrained MCU memory budgets.

### Secure Layer-2 Ethernet FOTA Bootloader

Compact secure update path operating over raw Ethernet frames without an IP stack. Used EC-PKI signature verification and direct MACPHY/SPI driver control to fit secure update behavior within a 32 KB bootloader region.

## Time-Triggered FreeRTOS Scheduler

Academic FreeRTOS v11.2.0 fork implementing static timeline scheduling, major/minor frame execution, and deadline violation hooks for safety-critical timing models on ARM Cortex-M3, validated under QEMU.

## EDUCATION

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### Politecnico di Torino – Turin, TO, Italy

Jan 2023 – Present

M.Sc. in Computer Engineering (part-time). Specialized in low level programming, safety-critical timing models, and automated synthesis/reliability verification of complex digital systems.

### University of Arkansas – Fayetteville, AR, USA

Aug 2021 – Dec 2021

International Exchange Program (SEC). Completed a full-time study load of in-person classes, laboratory sessions, and final exams.

### Politecnico di Torino – Turin, TO, Italy

Oct 2019 – Dec 2022

B.Sc. in Computer Engineering. Completed a rigorous, full-time study load combining core engineering principles (mathematics, physics, electronics) with a deep focus on computer science, advanced programming, operating systems, computer architectures, and network design.

## LANGUAGES, VISA AND GDPR

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Languages: English [C1 - Advanced], Arabic [Native], Italian [A2 - Basic].

Availability: Open to senior-level roles in Italy, EU relocation, hybrid, remote, or on-site engineering teams.

Work authorization: Permesso di Soggiorno lungo periodo (Italy / Long-term EU work authorization).

I declare that I have read the privacy statement on the processing of my personal data and my rights within the meaning of the General Data Protection Regulation (EU Regulation 2016/679) and Legislative Decree no. 196 of 30th June 2003: [didattica.polito.it/privacy/index\\_en.html](https://didattica.polito.it/privacy/index_en.html). I hereby give my consent to the processing of my personal data included in my CV.